# 1 Recap of probability theory

**(a) Outcomes, random variable and distribution.** $\Omega = \{HH, HT, TH, TT\}$, and $X$ counts heads:

| $\omega$ | $HH$ | $HT$ | $TH$ | $TT$ |
|---|---|---|---|---|
| $P(\{\omega\})$ | $1/4$ | $1/4$ | $1/4$ | $1/4$ |
| $X(\omega)$ | 2 | 1 | 1 | 0 |

$$p_X(0) = \tfrac{1}{4}, \qquad p_X(1) = \tfrac{1}{2}, \qquad p_X(2) = \tfrac{1}{4}.$$

**(b) Expectation values.**

$$\mathbb{E}[X] = 0 \cdot \tfrac{1}{4} + 1 \cdot \tfrac{1}{2} + 2 \cdot \tfrac{1}{4} = 1, \qquad \mathbb{E}[X^2] = 0^2 \cdot \tfrac{1}{4} + 1^2 \cdot \tfrac{1}{2} + 2^2 \cdot \tfrac{1}{4} = \tfrac{3}{2}.$$

$\mathbb{E}[X]$ is a fixed number, not a random variable.

**(c) Variance identity.** With the constant $\mu = \mathbb{E}[Z]$ (finite since $\mathbb{E}[Z^2] < \infty$), linearity gives

$$\begin{aligned}\mathbb{E}[(Z-\mu)^2] &= \mathbb{E}[Z^2 - 2\mu Z + \mu^2] \\ &= \mathbb{E}[Z^2] - 2\mu\mathbb{E}[Z] + \mu^2 = \mathbb{E}[Z^2] - \big(\mathbb{E}[Z]\big)^2.\end{aligned}$$

$$\boxed{\mathrm{Var}(X) = 3/2 - 1 = 1/2}.$$

**(d) Conditional probabilities.** $Y = 1$ on $HH, HT$ and $Y = 0$ on $TH, TT$:

| $x$ | Outcomes compatible with $X = x$ | $P(Y = 1 \mid X = x)$ | $\mathbb{E}[Y \mid X = x]$ |
|---|---|---|---|
| 0 | $TT$ | 0 | 0 |
| 1 | $HT, TH$ | $(1/4)/(1/2) = 1/2$ | $1/2$ |
| 2 | $HH$ | $(1/4)/(1/4) = 1$ | 1 |

Since $\mathbb{E}[Y \mid X = x] = 0 \cdot P(Y = 0 \mid X = x) + 1 \cdot P(Y = 1 \mid X = x)$, both columns agree, and both equal $x/2$.

# 2 Nadaraya–Watson regression

**(a) Reason about the estimator.** Let $S(x) = \sum_j K_h(x - x_j)$. For $S(x) > 0$: $w_i(x) \geq 0$ and $\sum_i w_i(x) = 1$.

(i) **One, no, or very many contributing observations.** Only observation $k$ contributes: $w_k(x) = 1$ and $\widehat{m}_h(x) = y_k$. None contributes: $S(x) = 0$, so the prediction is undefined (points at distance exactly $h$ do not contribute).

For $h \parallel \Delta$ (point spacing), about $2h/\Delta$ observations are averaged: less noise, but structure narrower than the window is smoothed away. For fixed $x$, as $h \to \infty$, $(1 - (x - x_i)^2/h^2)^2 \to 1$ and $15/(16h)$ cancels, so

$$w_i(x) \longrightarrow \frac{1}{N}, \qquad \widehat{m}_h(x) \longrightarrow \overline{y}.$$

(ii) **Reproducing the data versus predicting held-out data.** With $\Delta_{\min}$ the smallest adjacent gap, every $h \leq \Delta_{\min}$ isolates each observation, so $\widehat{m}_h(x_i) = y_i$ for all $i$, noise included.

Such $h$ predicts held-out observations poorly: the prediction is typically undefined, and slightly larger $h$ copies one noisy neighbor.

(iii) **Sampling density.** Denser sampling allows noise reduction at smaller $h$ and moves the onset of support gaps to smaller $h$. With $\Delta_{\max}$ the largest adjacent gap, $h > \Delta_{\max}/2$ suffices for positive support on the whole interval (at equality, a midpoint of a largest gap has no support). Every-fourth sampling roughly quadruples the gaps and the smallest usable $h$.

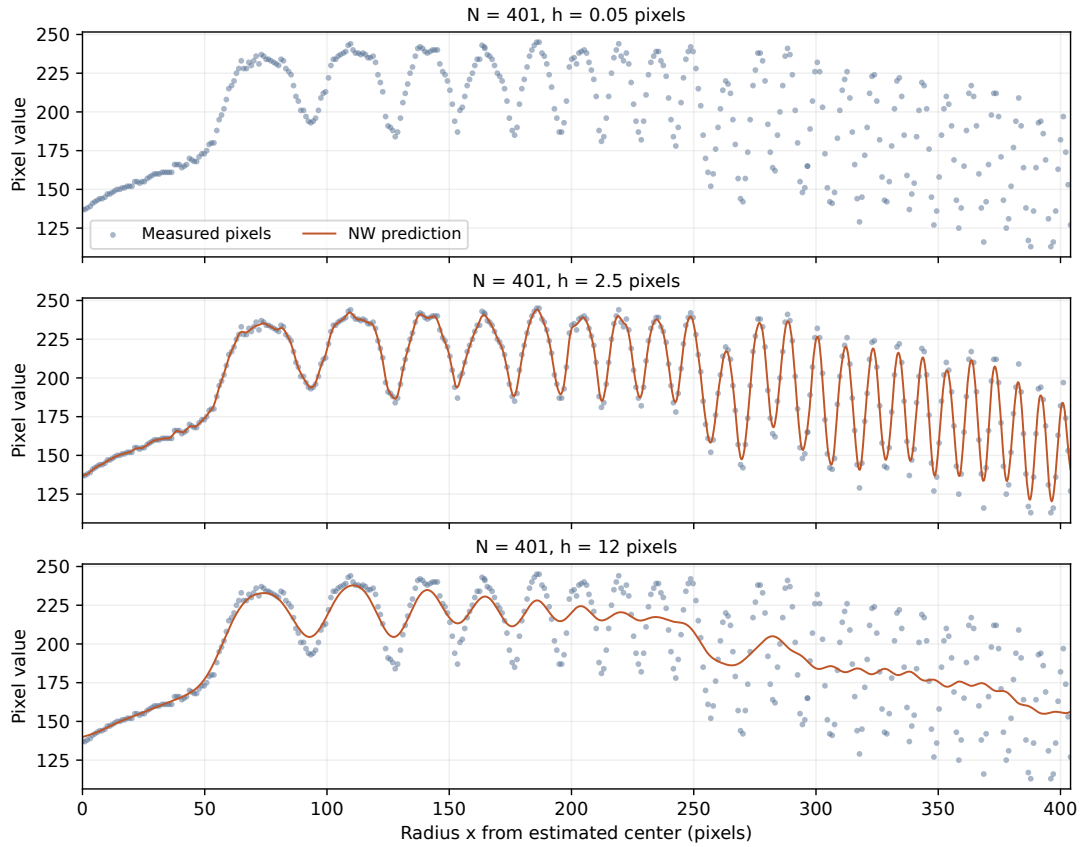
It does not resolve the fastest fringes: near the edge they are 9–10 px apart, while every fourth observation is about 4 px apart, i.e. only two to three samples per fringe. No $h$ recovers structure the samples do not resolve.

(iv) **Different oscillation rates and prediction range.** No: large $h$ attenuates the fast part, small $h$ averages less noise in the slow part, so one fixed $h$ must compromise.

Predictions cannot leave the observed range, since a defined prediction is a convex combination:

$$\min_{i:w_i(x)>0} y_i \;\le\; \widehat{m}_h(x) \;\le\; \max_{i:w_i(x)>0} y_i.$$

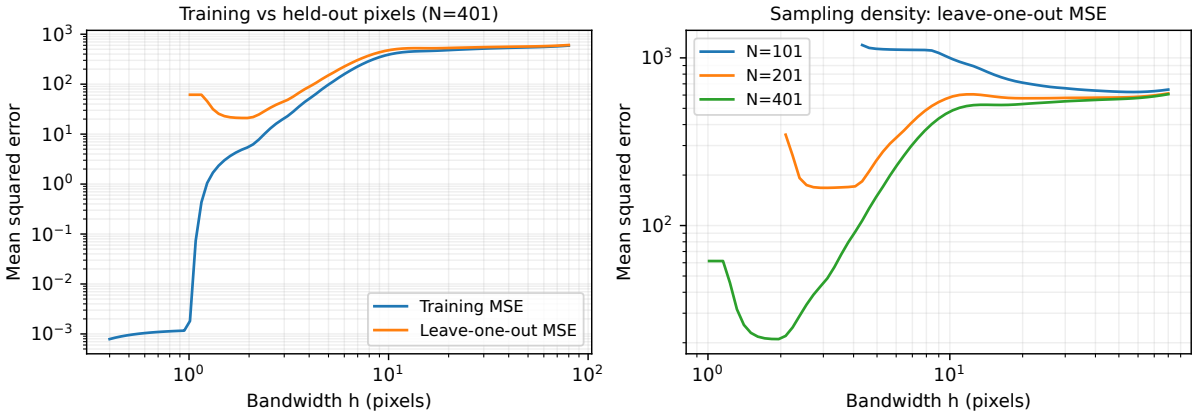
**(b) Implement and visualize.** Read `newton-rings.csv` (columns `x,y`) and use rows $0, 4, \ldots, 400$, $0, 2, \ldots, 400$, or all rows for $N = 101, 201, 401$. The kernel's normalization factor cancels in $w_i$. Mark undefined predictions ($S(x) = 0$) as NaN rather than zero.




Training error and leave-one-out (LOO) error, with observation $i$ omitted from numerator and denominator of $\widehat{m}_{h,-i}$:

$$T(h) = \frac{1}{N}\sum_i (\widehat{m}_h(x_i) - y_i)^2, \qquad C(h) = \frac{1}{N}\sum_i (\widehat{m}_{h,-i}(x_i) - y_i)^2.$$

$C$ is defined only if $h > \max_i \min_{j\neq i} |x_i - x_j|$, whereas $T(h) = 0$ for $h \le \Delta_{\min}$.




Grid: 81 logarithmically spaced $h$ from 0.4 to 80 px; zero training errors are omitted. Best tested bandwidths with full LOO support:

| $N$ | Max. gap (px) | Best tested $h$ (px) | LOO MSE |
|---|---|---|---|
| 101 | 4.123 | 57.448 | 625.575 |
| 201 | 2.146 | 3.117 | 167.827 |
| 401 | 1.162 | 1.961 | 21.026 |

**(c) Interpret the results.**

- Fits: $h = 0.05$ isolates observations and leaves undefined gaps between them. $h = 2.5$ has full coverage and preserves more fringes than $h = 12$, whose window spans several of the tightly spaced fringes and strongly attenuates them. The plots alone do not show which curve is closest to the unknown $f$.
- (a)(ii) confirmed: bandwidths with $T(h) = 0$ leave held-out points without support or predict them poorly.
- (a)(iii) confirmed: For $N = 201, 401$, $C(h)$ has a minimum which moves towards smaller $h$ for larger N as expected. For $N = 101$ no such minimum exists. Near the edge, fringes are 9–10 px apart and samples about 4 px apart, so averaging even immediate neighbors mixes bright and dark parts of a fringe. The best tested $h \approx 57$ px gives LOO MSE $\approx 626$, about half the subset's response variance ($\approx 1210$), i.e. little more than a slowly varying baseline. Adjusting $h$ cannot recover information lost through undersampling.
- (a)(iv) confirmed: at fixed $h$, attenuation is stronger in the fast part than in the slow part, so no single $h$ suits both. For example, with $N = 401$ and $h = 2.5$ px (figure in part (b)), the fit follows the fast fringes but overfits the slow part, showing small bumps from pixel-level fluctuations around $x \approx 35$–$50$. All fitted values stay within the measured range.

## 3 Linear regression

**(a) Visualize the data.** See the point clouds in the figure in part (g).

**(b) Inspect the data.** The first dataset lies close to a tilted plane; the second forms a well with an interior minimum. A plane suits only the first.

**(c) Derive the plane fit.**

$$X = \begin{pmatrix} 1 & x_{11} & x_{12} \\ \vdots & \vdots & \vdots \\ 1 & x_{N1} & x_{N2} \end{pmatrix}, \qquad \mathbf{y} = (y_1, \ldots, y_N)^\mathsf{T},$$

$$L(\boldsymbol{\beta}) = \mathbf{y}^\mathsf{T}\mathbf{y} - 2\boldsymbol{\beta}^\mathsf{T}X^\mathsf{T}\mathbf{y} + \boldsymbol{\beta}^\mathsf{T}X^\mathsf{T}X\boldsymbol{\beta}, \qquad \nabla L = 2X^\mathsf{T}(X\boldsymbol{\beta} - \mathbf{y}), \qquad \boxed{X^\mathsf{T}X\widehat{\boldsymbol{\beta}} = X^\mathsf{T}\mathbf{y}.}$$

The Hessian $2X^\mathsf{T}X$ is positive semidefinite, so every solution is a global minimizer. It is unique iff $\operatorname{rank}(X) = 3$, i.e. $\ker X = \{0\}$, i.e. the input points do not all lie on one affine line.

**(d) Fit the planes.** Least-squares fit with `numpy.linalg.lstsq`. All designs have rank three. Results, including part (g):

| Data | Model | Intercept | Coefficient 1 | Coefficient 2 | MSE |
|---|---|---|---|---|---|
| plane | plane | 1.006577 | 1.497626 | -0.795813 | 0.041070 |
| well | plane | 1.310940 | -0.044187 | -0.135601 | 2.210243 |
| well | squared | -0.990438 | 1.009798 | 0.686624 | 0.042929 |

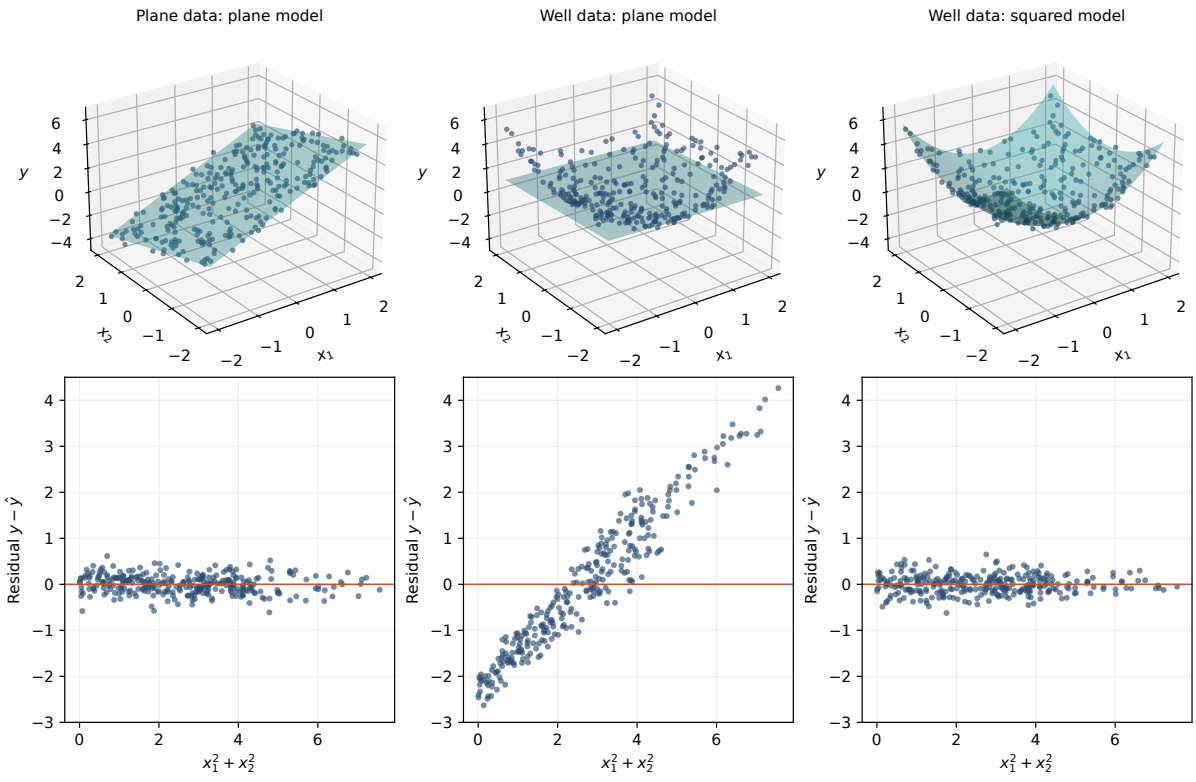
**(e) Assess the fits.** The plane on the well leaves a systematic pattern: mostly negative residuals near the center, positive farther out, with much larger MSE. This indicates missing curvature. The first plane shows no clear radial pattern. (The intercept enforces $\sum_i r_i = 0$, which does not by itself rule out structure.)

**(f) Set up the squared-feature model.**

$$\Phi = \begin{pmatrix} 1 & x_{11}^2 & x_{12}^2 \\ \vdots & \vdots & \vdots \\ 1 & x_{N1}^2 & x_{N2}^2 \end{pmatrix}.$$

**(g) Fit the well using squared features.** Solving $\Phi^\mathsf{T}\Phi\widehat{\boldsymbol{\theta}} = \Phi^\mathsf{T}\mathbf{y}$ with `lstsq` gives MSE 0.042929 and

$$\widehat{g}(x_1, x_2) = -0.990438 + 1.009798x_1^2 + 0.686624x_2^2.$$




Top: fitted surfaces and observations. Bottom: residuals against $x_1^2 + x_2^2$, on common vertical scales.

**(h) Compare and explain.** MSE drops from $2.210243$ (plane) to $0.042929$ with the same number of coefficients, and the radial residual trend disappears, leaving scatter consistent with noise. Both models are linear in their coefficients:

$$f_{\boldsymbol{\beta}}(\mathbf{x}) = (1, x_1, x_2)\boldsymbol{\beta}, \qquad g_{\boldsymbol{\theta}}(\mathbf{x}) = (1, x_1^2, x_2^2)\boldsymbol{\theta}.$$

The features are fixed before optimization. In the original inputs, the first model is affine and the second quadratic.